

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

This brainstorming session focuses on improving IT procurement by automating standard laptop orders using Flow Designer. The team can identify repetitive procurement activities, explore automation ideas, and collaborate on solutions that make standard laptop requests faster, more consistent, and easier to manage.

Use this template to identify automation opportunities, group related ideas, and prioritize practical solutions for standard laptop ordering through Flow Designer.

Reference: Flow Designer-based IT procurement automation

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement: Standard laptop procurement involves repetitive manual steps such as request submission, validation, approval, and order processing. The goal is to streamline these activities by using Flow Designer to automate the standard laptop ordering workflow.



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

- 🕒 10 minutes to prepare
- 🕒 1 hour to collaborate
- 👥 2-8 people recommended

➔

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

🕒 10 minutes

A Team gathering
Define who should participate in the session and send an invite. Share relevant information or pre-work ahead.

B Set the goal
Think about the problem you'll be focusing on solving in the brainstorming session.

C Learn how to use the facilitation tools
Use the Facilitation Superpowers to run a happy and productive session.

[Open article](#) ➔

1

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

🕒 5 minutes

PROBLEM

How might we [your problem statement]?



Key rules of brainstorming

To run a smooth and productive session

- 🗣️ Stay in topic.
- 💡 Encourage wild ideas.
- ⏸️ Defer judgment.
- 👂 Listen to others.
- 🗣️ Go for volume.
- 👁️ If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

- Automate standard laptop request submission and routing.
- Validate required request details automatically.
- Trigger approval workflows based on predefined conditions.
- Automatically create and assign procurement tasks.
- Send status notifications to requesters and procurement teams.
- Group ideas into Request Automation, Approval Automation, Task Automation, Notifications, and Tracking.

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

⌚ 10 minutes

TIP

You can select a sticky note and hit the pencil (switch to sketch) icon to start drawing!

Amar



Yuktesh



Person 3



Person 4



Person 5



Person 6



Person 7



Person 8



3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. In the last 10 minutes, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

⌚ 20 minutes

Person 4

TIP

Add customizable tags to sticky notes to make it easier to find, remove, organize, and categorize important ideas as themes within your mind.

Step-3: Idea Prioritization

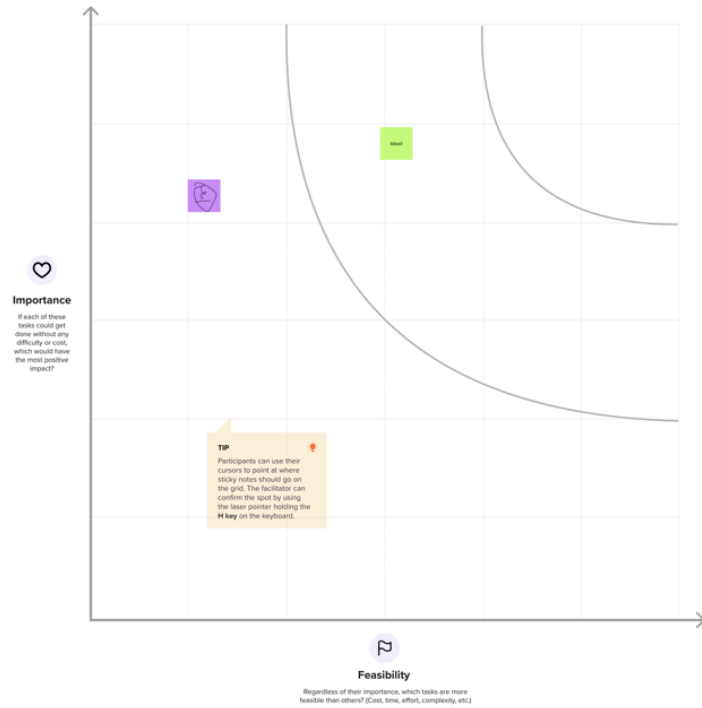
Prioritize ideas based on business value, ease of implementation, automation potential, time saved, consistency, and user experience. Give higher priority to solutions that can automate the complete standard laptop request-to-order process with minimal manual intervention.

4

Prioritize

Your team should all be on the same page about what's important moving forward. Place your ideas on this grid to determine which ideas are important and which are feasible.

🕒 20 minutes



Ideation Phase

Define the Problem Statements

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Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - <i>who are they?</i>	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - <i>what are they trying to achieve?</i>	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way - <i>what bothers them most?</i>	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists - <i>what needs to be solved?</i>	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view - <i>how does it impact them emotionally?</i>	Describe the emotions the result from experiencing the problems or barriers

Reference: Customer Problem Statement

Problem Statements:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1					
PS-2					

Ideation Phase

Empathize & Discover

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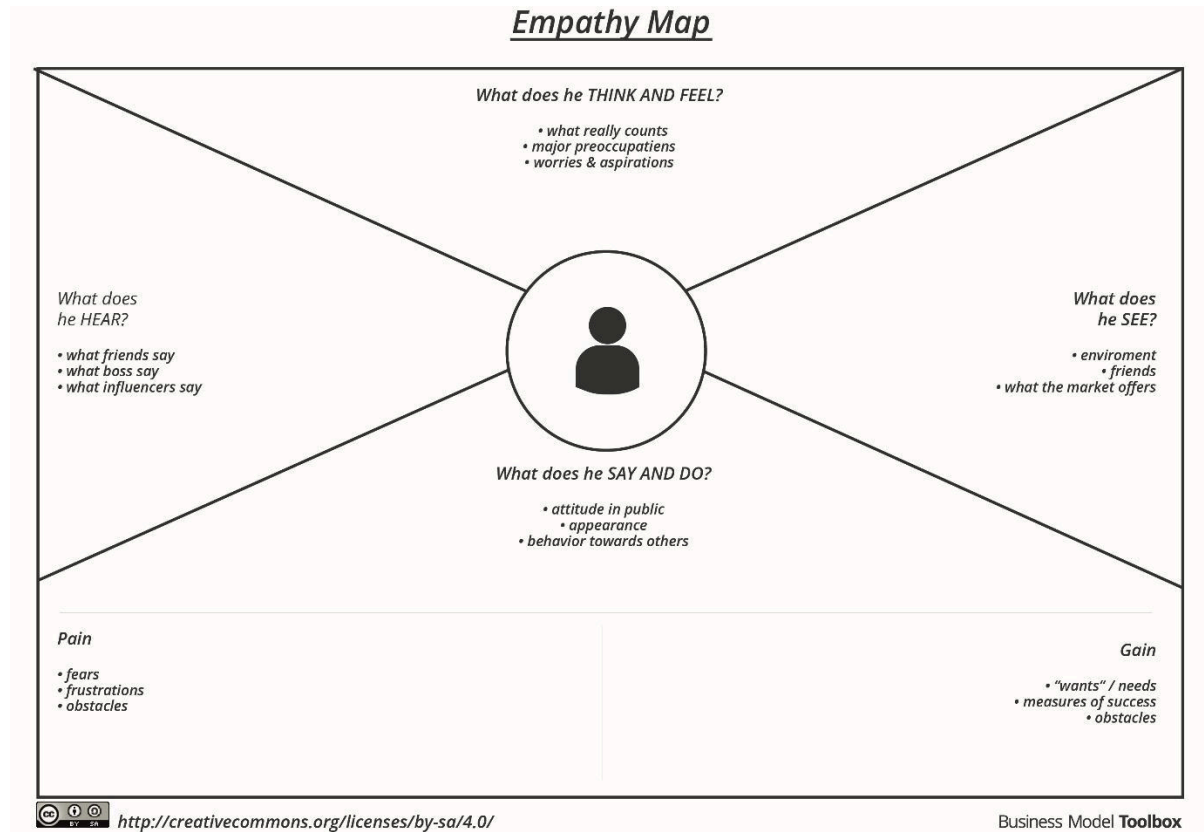
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Reference: Empathy Map Canvas

Example: Standard Laptop Procurement & Ordering Workflow

User / Customer

Employee requesting a standard laptop and the IT procurement team processing the request.

SAYS

- "I want to request a standard laptop easily."
- "I want to know the status of my request."
- "I don't want to follow up repeatedly for approvals."

THINKS

- The procurement process should be simple and quick.
- The request should automatically move to the correct approval stage.
- Manual steps may cause delays or missed updates.

DOES

- Submits laptop requests.
- Provides required request details.
- Follow up on approvals and order status.
- The procurement team reviews requests and performs manual tasks.

FEELS

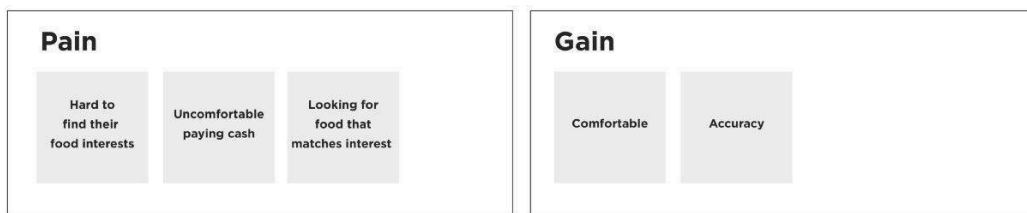
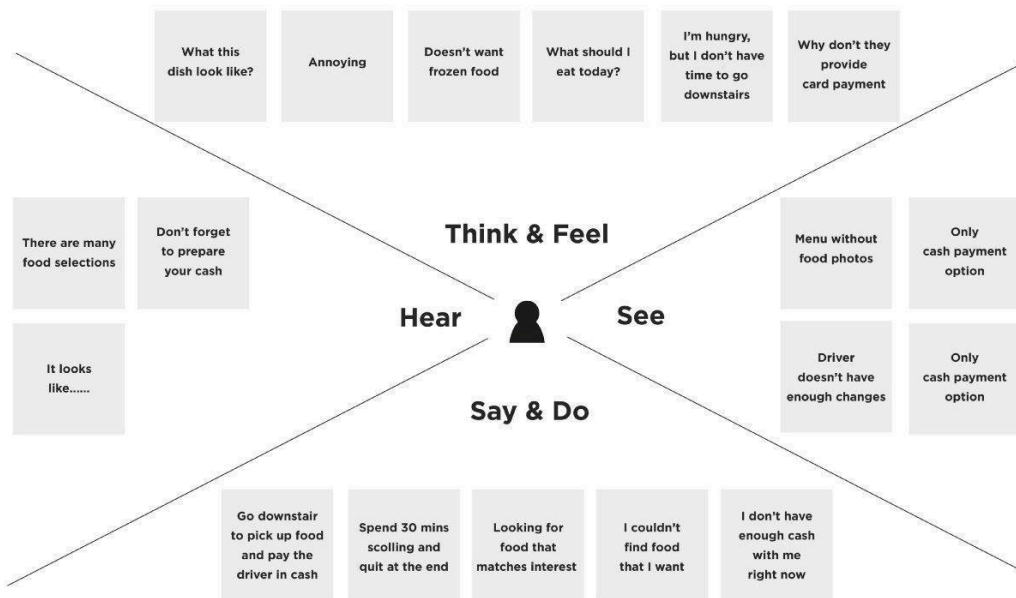
- Frustrated by repetitive manual steps.
- Concerned about delays in approval and ordering.
- I want a transparent and easy-to-track process.

PAINS

- Multiple manual activities.
- Delays in approvals and task assignment.
- Difficulty tracking request status.
- Repeated follow-ups between teams.

GAINS

- Faster laptop request processing.
- Automated validation and approvals.
- Automatic task creation and assignment.
- Clear status notifications and tracking.
- Reduced manual effort through Flow Designer.



UX JOURNEY MAP

Standard Laptop Procurement Journey

Automating standard laptop orders with Flow Designer



Project Design Phase-II

Data Flow Diagram & User Stories

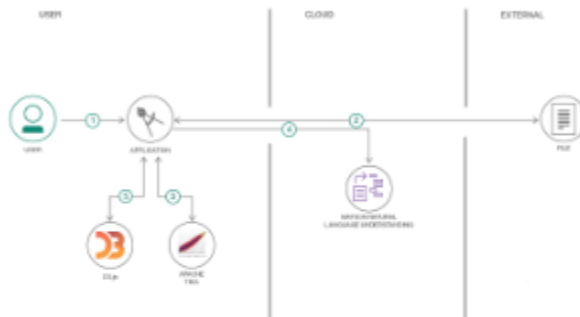
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Data Flow Diagrams:

A Data Flow Diagram (DFD) for the proposed solution represents how a standard laptop order moves through the IT procurement process using Flow Designer. It shows how an employee submits a laptop request, how the request is validated and routed for approval, how the IT procurement team processes the approved order, and how status notifications and order information are stored and updated.

Example: Standard Laptop Order Flow (Simplified)([Simplified](#))

Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the Standard Laptop Order Automation product.

<u>Requester (Employee)</u>	<u>Laptop Request</u>	<u>USN-1</u>	<u>As an employee, I can submit a standard laptop order request through the procurement request form.</u>	<u>I can submit the required laptop details and receive a request number.</u>	<u>High</u>	<u>Sprint-1</u>
	<u>Request Validation</u>	<u>USN-2</u>	<u>As an employee, I can receive a confirmation when my laptop request is submitted and validated.</u>	<u>I can receive confirmation with the request status and request number.</u>	<u>High</u>	<u>Sprint-1</u>
	<u>Approval</u>	<u>USN-3</u>	<u>As an employee, I can track the approval status of my standard laptop order request.</u>	<u>I can view whether the request is pending, approved, or rejected.</u>	<u>High</u>	<u>Sprint-1</u>
	<u>Order Processing</u>	<u>USN-4</u>	<u>As an IT procurement team member, I can process an approved standard laptop order automatically through Flow Designer.</u>	<u>An approved request creates the required procurement task and routes it to the appropriate team.</u>	<u>High</u>	<u>Sprint-2</u>
	<u>Notifications</u>	<u>USN-5</u>	<u>As a requester, I can receive notifications when my laptop order changes status.</u>	<u>I can receive notifications for approval, processing, and completion of the order.</u>	<u>Medium</u>	<u>Sprint-2</u>
	<u>Dashboard</u>					
<u>IT Procurement Team</u>						
<u>Approver</u>						
<u>Administrator</u>						

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Project Design Phase-II

Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Laptop Request Submission	Employee submits a standard laptop request through the procurement request form.
FR-2	Request Validation & Approval	System validates request details and routes the request for required approval.
FR-3	Automated Laptop Order Creation	Flow Designer automatically creates and processes the standard laptop procurement order after approval.
FR-4	Order Tracking & Notifications	Requester receives status notifications and can track the laptop order through completion.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple and user-friendly laptop request process with clear request status.
NFR-2	Security	Ensure only authorized users can submit, approve, and access procurement requests.
NFR-3	Reliability	Ensure automated flows process approved standard laptop orders consistently without data loss.

NFR-4	Performance	Process requests and trigger order workflows efficiently with minimal delay.
NFR-5	Availability	Keep the procurement request and order tracking process available during business operations.
NFR-6	Scalability	Support increasing numbers of laptop requests and procurement transactions without major redesign.

Project Design Phase-II Technology Stack (Architecture & Stack)

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Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Reference: ServiceNow Flow Designer documentation

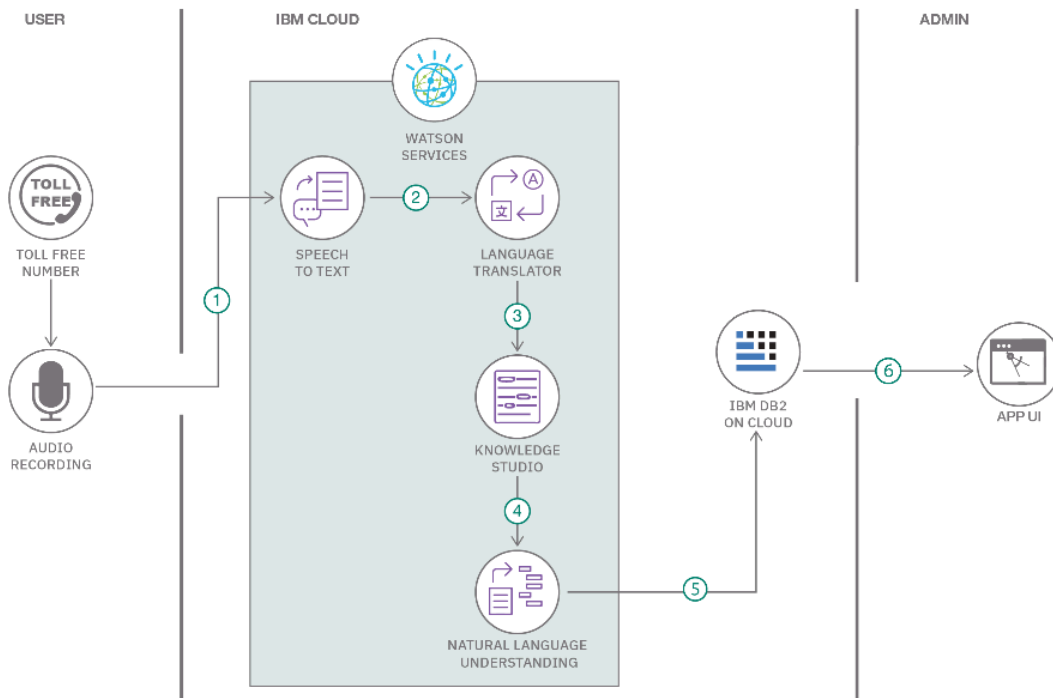


Table-1 : Components & Technologies:

S. No	Component	Description	Technology
1	User Interface	Employee laptop request form and approval interface	ServiceNow Service Catalog / Portal
2	Application Logic-1	Validates standard laptop request details and triggers the procurement workflow	ServiceNow Flow Designer
3	Application Logic-2	Routes the request for manager/IT approval and updates request status	ServiceNow Flow Designer
4	Application Logic-3	Creates procurement and fulfillment tasks after approval	ServiceNow Flow Designer / ServiceNow ITSM
5	Database	Stores laptop request, requester, approval and fulfillment information	ServiceNow tables / CMDB
6	Cloud Database	Cloud-based platform data storage for procurement workflow records	ServiceNow Cloud Platform
7	File Storage	Stores supporting documents such as quotations or procurement attachments	ServiceNow Attachments
8	External API-1	Optional integration with supplier/procurement system for laptop availability and order details	REST API
9	External API-2	Optional integration for shipment or delivery status updates	REST API
10	Machine Learning Model	Not required for standard laptop order automation	Not Applicable
11	Infrastructure (Server / Cloud)	Cloud deployment of the procurement request and workflow automation	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source framework is required for the core workflow	ServiceNow Flow Designer
2	Security Implementations	Role-based access, authentication, authorization and controlled procurement approvals	ServiceNow ACLs / Roles
3	Scalable Architecture	Workflow automation can handle multiple laptop requests using reusable flows and subflows	ServiceNow Flow Designer
4	Availability	Cloud-hosted workflow platform supports continuous access to request and approval processes	ServiceNow Cloud Platform
5	Performance	Automated routing, approvals and task creation reduce manual processing time	Flow Designer / ServiceNow Platform

References:

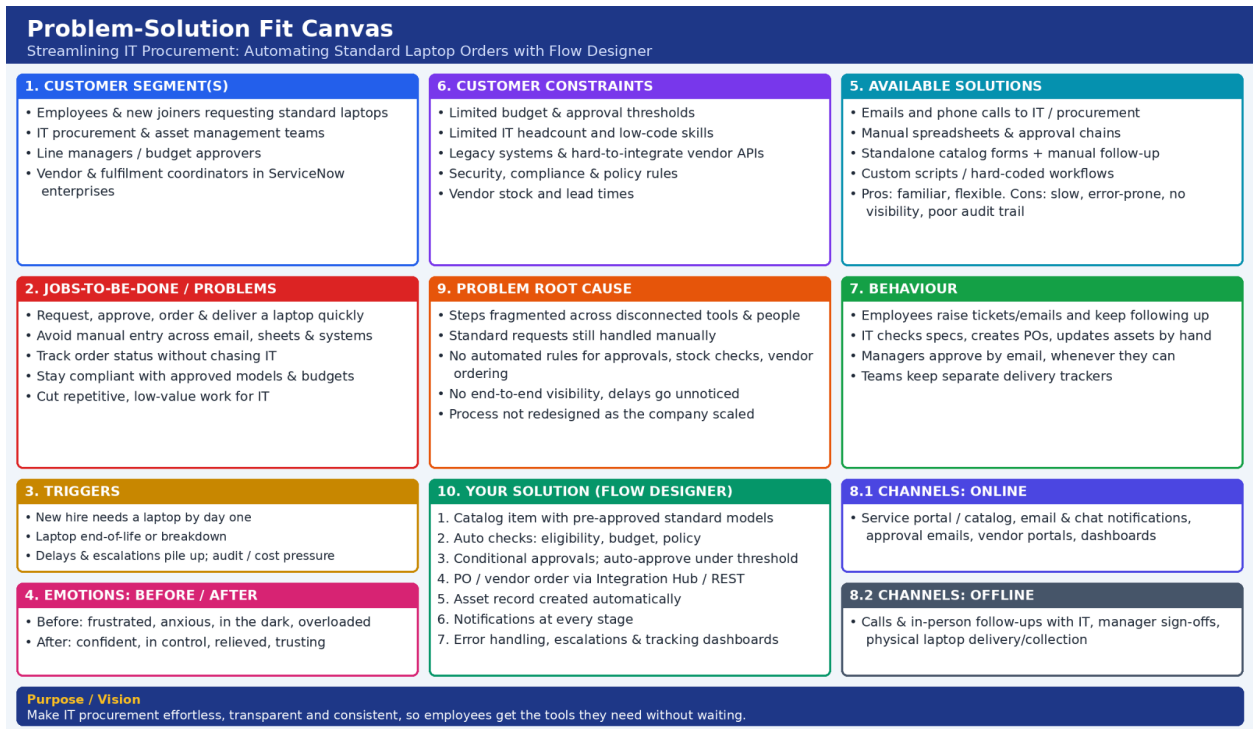
<https://c4model.com/>

<https://www.servicenow.com/products/flow-designer.html>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>



PROJECT DESIGN PHASE

PROBLEM – SOLUTION FIT PROJECT REPORT

Automating Standard Laptop Procurement with Flow Designer

Project Report

Project Name	Automating Standard Laptop Procurement with Flow Designer
Technology	ServiceNow – Flow Designer
Project Focus	Problem–Solution Fit for IT Procurement Automation

1. INTRODUCTION

1.1 Project Overview

This project focuses on automating the procurement process for standard laptops using ServiceNow Flow Designer. The solution provides a structured workflow for submitting a laptop request, validating the information, routing the request for approval, creating procurement or fulfillment tasks, sending notifications, and tracking the request until completion.

1.2 Purpose

The purpose is to reduce repetitive manual coordination, minimize delays caused by email-based communication, standardize approvals, improve request visibility, and maintain an auditable procurement record.

2. IDEATION PHASE

2.1 Problem Statement

Standard laptop procurement can involve several manual activities, including collecting requester information, checking the selected device, obtaining approval, coordinating with IT or procurement teams, and updating the requester. These activities may result in incomplete requests, inconsistent routing, duplicate communication, and limited status visibility.

2.2 Customer Segment

Primary customers are employees or staff members who require a standard company laptop, along with IT support and procurement teams responsible for approving and fulfilling the request.

2.3 Jobs-to-be-Done / Problems

- Submit a standard laptop request with complete information.

- Obtain the required approval through a defined process.

- Validate that the selected laptop belongs to the approved standard options.

- Coordinate procurement and fulfillment without repeated manual follow-ups.

- Track request progress from submission to completion.

2.4 Customer Behaviour

The requester submits laptop details and responds to clarification requests when required. IT and procurement teams review information, approve or reject requests, perform fulfillment activities, and update the request status.

2.5 Triggers

A new employee requires a standard laptop.

An existing employee needs a replacement device.

A device becomes unavailable or requires replacement.

A business team submits a standard equipment requirement.

2.6 Emotions: Before / After

Before automation, the requester may feel uncertain because the approval and fulfillment status is not immediately visible. After automation, the requester receives structured notifications and can follow the request through defined stages.

3. REQUIREMENT ANALYSIS

3.1 Available Solutions

Email-based procurement requests.

Manual coordination between employees, IT, managers, and procurement.

Spreadsheets or manually maintained request records.

ServiceNow catalog/request capabilities with Flow Designer automation.

3.2 Customer Constraints

Incomplete requester or delivery information.

Limited visibility into request status.

Different approval requirements for different requests.

Non-standard laptop requirements that need manual review.

Integration limitations with inventory, vendor, or finance systems.

3.3 Problem Root Cause

The main root cause is the dependence on manual, disconnected steps for request collection, validation, approval, communication, and fulfillment. Without a centralized workflow, information can be incomplete or repeatedly communicated, while status updates depend on manual follow-up.

3.4 Channels of Behaviour

Online channels include the ServiceNow request/catalog interface, automated notifications, and request status records. Offline activities include IT or procurement review, device preparation, and physical laptop handover.

4. PROJECT DESIGN

4.1 Proposed Solution

The proposed solution is a Flow Designer-based workflow for standard laptop procurement. The requester selects an approved laptop model and enters the required information. Flow Designer validates the request, determines the approval path, sends the approval request, creates fulfillment tasks after approval, sends notifications, and updates the request status. Rejected, incomplete, or non-standard requests can be routed through exception handling or manual review.

4.2 Solution Requirements

- Capture requester and delivery information.

- Provide a controlled list of standard laptop models.

- Validate mandatory information.

- Route requests to the appropriate approver.

- Create procurement or fulfillment tasks.

- Send automated notifications.

- Maintain request status and history.

- Support rejection and exception paths.

- Maintain an auditable workflow record.

- Use role-based access and maintain traceability.

4.3 Solution Architecture

The solution contains five logical layers: Request Layer, Data and Validation Layer, Automation Layer, Approval and Fulfillment Layer, and Notification/Reporting Layer. Flow Designer acts as the central orchestration component connecting these layers.

5. CUSTOMER JOURNEY / WORKFLOW

Requester identifies the need for a standard laptop.

Requester submits the procurement request.

The system validates mandatory information and the selected standard model.

Request is routed to the appropriate approver.

Requester receives approval or rejection notification.

Approved request generates a procurement or fulfillment task.

IT/procurement updates the fulfillment status.

The request is closed and completion is communicated to the requester.

6. TECHNOLOGY STACK

Primary Platform: ServiceNow

Workflow Automation: Flow Designer

Request Intake: ServiceNow Catalog / Request capabilities

Notifications: ServiceNow notification mechanisms

Access Control: ServiceNow roles and permissions

Reporting & Monitoring: ServiceNow reporting and workflow execution history

7. PROJECT PLANNING & SCHEDULING

Analyze the existing procurement process and identify manual steps.

Define the standard laptop catalog and business rules.

Configure request fields and validation.

Design and build the Flow Designer workflow.

Configure approval, notifications, and fulfillment tasks.

Perform functional and performance testing.

Document the solution and prepare deployment.

Monitor the workflow and refine rules based on operational feedback.

8. FUNCTIONAL AND PERFORMANCE TESTING

Testing should cover valid submissions, missing mandatory information, invalid laptop selections, approval, rejection, fulfillment task creation, notification delivery, and exception handling. Performance testing should consider workflow execution time, request responsiveness, task creation latency, and behaviour under increased request volume.

9. ADVANTAGES & LIMITATIONS

9.1 Advantages

Reduces repetitive manual coordination.

Standardizes laptop request information.

Automates approval routing and notifications.

Improves request status visibility.

Creates a consistent audit trail.

Supports measurable procurement and fulfillment stages.

Provides a foundation for future IT procurement automation.

9.2 Limitations

Initial configuration requires process analysis and testing.

Incorrect business rules can automate an incorrect process.

Non-standard requirements may still require manual review.

Inventory, vendor, or finance integrations may require additional configuration.

Approval-policy or laptop-standard changes require controlled maintenance.

10. CONCLUSION

Automating Standard Laptop Procurement with Flow Designer provides a structured approach to managing IT equipment requests. By combining standardized request data, validation, approvals, notifications, and fulfillment tasks in one workflow, the solution addresses repetitive manual coordination and improves process transparency. The same approach can later be extended to other standardized IT equipment and services.

11. FUTURE SCOPE

Automated inventory availability checks.

Integration with asset management and vendor systems.

Automated purchase order creation.

Cost-center validation.

SLA monitoring and escalation rules.

Approval delegation.

Analytics dashboards.

Automated asset assignment after delivery.

Support for monitors, mobile devices, and peripherals.

12. APPENDIX

12.1 Source Code / Configuration

The implementation is intended to be configuration-driven in ServiceNow Flow Designer. Exported flows, update sets, configuration notes, scripts, and integration definitions may be attached to the final project package where applicable.

12.2 Dataset

No external dataset is required for the basic workflow. Representative requester, laptop model, approval, and fulfillment records can be used as test data.

12.3 References

The Problem–Solution Fit structure is based on the provided Problem–Solution Fit Template and sample canvas, which organize the analysis around customer segments, jobs/problems, behaviour, triggers, emotions, channels, constraints, root cause, available solutions, and the proposed solution.

Project Design Phase

Proposed Solution Template

Date: 01 October 2026

Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Maximum Marks: 2 Marks

Proposed Solution Template:

The project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	The current IT laptop procurement process can involve multiple manual steps such as employee requests, approval, verification, purchase order creation, and order tracking. These manual activities can cause delays, repeated data entry, communication gaps, and difficulty in tracking request status. A streamlined and automated process is needed to make standard laptop procurement faster, consistent, and easier to manage.
2	Idea / Solution description	The proposed solution uses Flow Designer to automate the standard laptop procurement workflow. When an employee submits a laptop request, the flow validates the request, routes it to the appropriate approver, and initiates the required procurement steps after approval. Notifications and status updates can also be automated so that employees and IT/procurement teams can track the request throughout the process. This reduces manual effort and creates a structured end-to-end workflow.

3	Novelty / Uniqueness	The solution combines request management, approval automation, procurement workflow, notifications, and status tracking in a single automated flow. Instead of handling every standard laptop request manually, predefined rules can automatically manage repetitive tasks. This provides a consistent process while reducing unnecessary human intervention.
4	Social Impact / Customer Satisfaction	The solution improves the experience of employees by reducing waiting time and providing better visibility into their laptop requests. IT and procurement teams can spend less time on repetitive administrative tasks and more time on important activities. Automated notifications and clear request statuses also improve communication and transparency between employees, IT, and procurement teams.
5	Business Model (Revenue Model)	The solution can be implemented as an internal IT process automation system within organizations. It can reduce operational costs by minimizing manual processing, reducing administrative effort, and improving resource utilization. For organizations offering workflow automation services, the solution can also be provided as a customized implementation or automation service.
6	Scalability of the Solution	The workflow can be expanded beyond standard laptop orders to automate other IT equipment requests such as monitors, keyboards, docking stations, and other approved hardware. Additional approval rules, departments, locations, vendors, and procurement conditions can be incorporated as the organization grows. The same Flow Designer approach can therefore support a wider range of IT procurement processes.

Project Design Phase

Solution Architecture

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Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

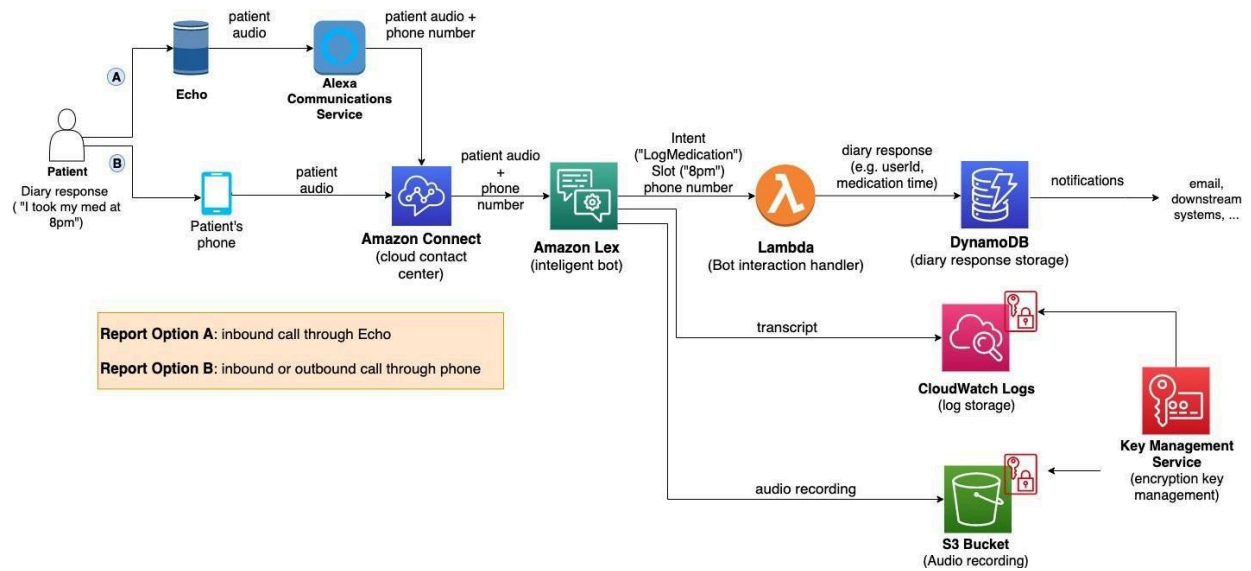
- Find the best tech solution to solve existing business problems.

- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.

- Define features, development phases, and solution requirements.

- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:



Reference:

<https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

A Sprint

A fixed period or duration in which a team works to complete a set of tasks.

An Epic

A big task or project that is too large to complete in one sprint. It is broken down into smaller tasks (stories) that can be completed over multiple sprints.

A Story

A small task. It is part of an Epic.

A Story Point

A number that represents how much effort a story takes to complete (usually in the form of a Fibonacci series).

1 - Very Easy task

2 - Normal task

3 - Moderate task

5 - Difficult task

Storypoint - 5 (1, 2, 3, 5)

Sprint 1

Laptop Request Intake & Validation (Epic 1)

Creating Laptop Request Form (USN1) 2

Capturing Employee & Department Details (USN2) 1

Validating Standard Laptop Eligibility (USN3) 3

Generating Request Number (USN4) 2

Total Story Point in Sprint 1 = 2+1+3+2 = 8

Sprint 2

Automated Procurement Workflow (Epic 2)

Manager Approval Flow (USN5) 3

Procurement Team Notification (USN6) 2

Automatic Purchase Order Creation (USN7) 5

Laptop Inventory & Delivery Update (USN8) 3

Total Story Point in Sprint 2 = 3+2+5+3 = 13

Sprint 3

Flow Designer Monitoring & Closure (Epic 3)

Tracking Request Status (USN9) 2

Automated Employee Notification (USN10) 2

Request Closure & Record Update (USN11) 3

Total Story Point in Sprint 3 = 2+2+3 = 7

Total Story Points

Sprint 1 = 8

Sprint 2 = 13

Sprint 3 = 7

Velocity = Total Story Points Completed / Number of Sprints

Total Story Points = 8+13+7 = 28

No. of Sprints = 3

Velocity = 28/3

9.33 (Story Points per Sprint)

Your team's velocity is 9.33 Story Points per Sprint.

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

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Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Laptop Request Intake & Validation	USN-1	As an employee, I can submit a standard laptop request through a request form.	2	High	
Sprint-1	Laptop Request Intake & Validation	USN-2	As a requester, I can provide employee, department and laptop requirement details.	1	High	
Sprint-1	Laptop Request Intake & Validation	USN-3	As the system, I can validate whether the request meets standard laptop eligibility criteria.	3	High	
Sprint-1	Laptop Request Intake & Validation	USN-4	As the system, I can generate a request number for tracking the laptop order.	2	Medium	

Sprint-2	Automated Procurement Workflow	USN-5	As a manager, I can approve or reject a standard laptop request through the workflow.	3	High	
Sprint-2	Automated Procurement Workflow	USN-6	As the procurement team, I can receive an automatic notification after approval.	2	High	
Sprint-2	Automated Procurement Workflow	USN-7	As the system, I can automatically create the procurement/purchase order record.	5	High	
Sprint-2	Automated Procurement Workflow	USN-8	As the system, I can update laptop inventory and delivery information.	3	Medium	
Sprint-3	Flow Designer Monitoring & Closure	USN-9	As an IT user, I can track the status of the laptop procurement request.	2	Medium	
Sprint-3	Flow Designer Monitoring & Closure	USN-10	As an employee, I receive automatic notifications about request progress.	2	Medium	
Sprint-3	Flow Designer Monitoring & Closure	USN-11	As the system, I can close the request after successful laptop delivery and update the record.	3	High	

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	6 Days	01 Oct 2026	06 Oct 2026	8	06 Oct 2026
Sprint-2	13	6 Days	07 Oct 2026	12 Oct 2026	13	12 Oct 2026
Sprint-3	7	6 Days	13 Oct 2026	18 Oct 2026	7	18 Oct 2026

Velocity:

Total Story Points = $8 + 13 + 7 = 28$

Number of Sprints = 3

Average Velocity (AV) = $28 / 3 = 9.33$ Story Points per Sprint

Average Velocity per Day = $9.33 / 6 = 1.56$ Story Points per Day

Burndown Chart:

A burndown chart is a graphical representation of work left to do versus time. For this project, it can be used to track the remaining story points across the three sprints as the automated laptop procurement workflow is developed and completed.

References:

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Project Development Phase

Flow Performance Test

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	

Flow Performance Testing:

The project team shall fill the following information in flow performance testing template.

S.No.	Parameter	Values	Screenshot
1	Flow Summary	Flow - Standard Laptop Order. Trigger - Service Catalog. Actions - Ask for Approval, Create Catalog Task, Update Record, Send Email.	
2	Flow Accuracy	Execution Success Rate - 100% (10 of 10 test requests) Approval Routing Accuracy - 100%	
3	Flow Optimization Result (if Done)	Avg. Order Processing Time - under 2 minutes (after adding conditional logic and retry on action failure)	

Functional & Performance Testing – IT Procurement Automation

Flow Designer Functional & Performance Test

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Laptop Request Input Validation	Enter valid and invalid laptop request details such as laptop type, requester details and business justification	Valid request details are accepted and invalid or missing details show validation errors		
FT-02	Quantity Input Validation	Enter valid and invalid laptop quantities	Accepts valid quantities and shows an error for invalid or out-of-range values		
FT-03	Standard Laptop Order Creation	Submit a complete standard laptop request through the ServiceNow catalog	A standard laptop procurement request is created with the correct request details		
FT-04	Procurement Integration Check	Check the configured supplier/procurement integration and verify that the workflow can exchange order information	The integration responds successfully and required procurement information is transferred		

PT-01	Workflow Response Time Test	Measure the time taken from submitting a laptop request to workflow initiation	Workflow should initiate within the defined response-time target		
PT-02	Workflow Processing Load Test	Submit multiple standard laptop requests at the same time	Requests should be processed without significant workflow delay or failure		
PT-03	Procurement Attachment Load Test	Upload multiple procurement documents or quotations and verify workflow processing	Attachments should be processed smoothly without workflow failure		

Project Development Phase

Flow Performance Test

Date	10 February 2025
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	10 Marks

Flow Performance Testing:

The project team shall fill the following information in flow performance testing template.

S.No.	Parameter	Values	Screenshot
1	Metrics	Flow Execution Metrics: Total Test Requests - 10, Successful Executions - 10, Failed Executions - 0, Success Rate - 100% Fulfilment Metrics: Avg. Order Processing Time - under 2 minutes, Approval Routing Accuracy - 100% & Catalog Task Creation - Passed	
2	Optimize the Flow	Flow Optimization - Conditional logic for standard laptop models, retry on action failure Validation Method - Flow Designer Test feature & UAT using catalog requests	

Project Development Phase Performance Test

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	

Flow Performance Testing:

The project team shall fill the following information in the flow performance testing template.

S.No.	Parameter	Screenshot / Values
1	Data Rendered	Catalog request data (laptop model, requester, department, cost center) captured through the Service Catalog trigger and shown as data pills.
2	Data Preprocessing	Requester details auto-fetched; laptop model validated against the standard laptop list; mandatory fields checked before the flow proceeds.
3	Utilization of Flow Conditions	Condition - Catalog Item is Standard Laptop; Look Up Record filter - Active is true; approval required only when cost exceeds the limit.
4	Data Pills & Actions Used	Trigger data pills, Ask for Approval, Create Catalog Task (procurement), Update Record (request item state), Send Email notification.
5	Dashboard design	No of Visualizations / Graphs - 4 (Requests by status, Avg. fulfilment time, Orders by laptop model, Pending approvals)
6	Report Design	No of Visualizations / Graphs - 3 (Monthly laptop orders, Approval turnaround time, Successful vs failed flow executions)

Project Development Phase

Flow Performance Test

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	

Flow Performance Testing:

The project team shall fill the following information in flow performance testing template.

S.No.	Parameter	Values	Screenshot
1	Flow Summary	ServiceNow Flow Designer automation for standard laptop orders using Service Catalog, Approvals and Catalog Tasks. Note : If the request matches the standard laptop list then the flow continues and the procurement task is created, or else it will show an error.	
2	Flow Accuracy	Execution Success Rate - 100% Approval Routing Accuracy - 100%	
3	Flow Validation Result	Condition Check - Request continues only if the laptop model matches the standard laptop list Error Handling - If a mandatory field is blank or	

		the model is invalid, the flow stops and shows an error	
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Project Development Phase

Performance Test

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	

Flow Performance Testing:

The project team shall fill the following information in flow performance testing template.

S.No.	Parameter	Screenshot / Values
1	Data Rendered	Catalog request data (laptop model, requester, department, cost center) captured through the Service Catalog trigger and shown as data pills.
2	Data Preprocessing	Requester details auto-fetched; laptop model validated against the standard laptop list; mandatory fields checked before the flow proceeds.
3	Utilization of Flow Conditions	Condition - Catalog Item is Standard Laptop; Look Up Record filter - Active is true; approval required only when cost exceeds the limit.
4	Data Pills & Actions Used	Trigger data pills, Ask for Approval, Create Catalog Task (procurement), Update Record (request item state), Send Email notification.
5	Dashboard design	No of Visualizations / Graphs - 4 (Requests by status, Avg. fulfilment time, Orders by laptop model, Pending approvals)
6	Report Design	No of Visualizations / Graphs - 3 (Monthly laptop orders, Approval turnaround time, Successful vs failed flow executions)

User Acceptance Testing (UAT) Template

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	

Project Overview:

Project Name: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Project Description: A ServiceNow Flow Designer flow that automates standard laptop orders, from the catalog request through approval, procurement task creation and requester notification.

Project Version: 1.0

Testing Period: [Start Date] to [End Date]

Testing Scope:

Features: Standard Laptop catalog item, approval routing, procurement task creation, email notifications and request status updates.

User Stories: As an employee, I can order a standard laptop from the Service Catalog. As a manager, I can approve or reject the request. As a procurement team member, I receive a task automatically after approval.

Testing Environment:

URL/Location: ServiceNow instance - <https://<your-instance>.service-now.com> (Service Catalog and Flow Designer)

Credentials (if required): Test user accounts for employee, manager and procurement roles (shared by the project team separately)

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Order a standard laptop from the Service Catalog	Log in as an employee Open Standard Laptop item in the catalog Fill required fields and submit	Request and request item are created and the flow is triggered	[Record the actual outcome]	[Pass/Fail]
TC-002	Manager approves the laptop request	Log in as the approving manager Open the pending approval Click Approve	Flow moves to procurement and a catalog task is created	[Record the actual outcome]	[Pass/Fail]
TC-003	Manager rejects the request (negative)	Log in as the approving manager Open the pending approval Click Reject and add a comment	Request is closed as rejected and the requester gets an email	[Record the actual outcome]	[Pass/Fail]

TC-004	Submit without mandatory fields (negative)	Open Standard Laptop item Leave a mandatory field blank Click Order Now	Submission is blocked with a validation message	[Record the actual outcome]	[Pass/Fail]
TC-005	Complete the procurement task	Log in as a procurement team member Open the assigned catalog task Close the task as complete	Request is marked Closed Complete and the requester is notified	[Record the actual outcome]	[Pass/Fail]
TC-006	Verify the flow execution	Open Flow Designer and go to Executions Find the run for the test request Review the status of each action	All actions show Completed with no errors	[Record the actual outcome]	[Pass/Fail]

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Approval email not received by the manager	Submit a standard laptop request Check the manager's inbox No email is received	Medium	Open	Sample entry - check the Send Email action and notification settings
BG-002	Request stays Open after the task is closed	Complete the procurement task Open the request item State is not updated	High	In Progress	Sample entry - replace with your actual findings

Sign-off:**Tester Name:** [Name of Tester]**Date:** [Date of Test Completion]**Signature:** [Tester's Signature]**Notes:**

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement.

- **Bug tracking should include details such as severity, status, and steps to reproduce.**
- **Obtain sign-off from both the project manager and product owner before proceeding with deployment.**

Acceptance Testing

UAT Execution & Report Submission

Date	01 October 2026
Team ID	SWTID-2026-7461
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer project at the time of the release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	1	0	1	0	2
Duplicate	0	1	0	0	1
External	0	0	1	0	1
Fixed	2	3	2	1	8
Not Reproduced	0	0	1	0	1
Skipped	0	0	0	1	1
Won't Fix	0	1	1	0	2
Totals	3	5	6	2	16

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested

Section	Total Cases	Not Tested	Fail	Pass
Catalog Item Submission	6	0	0	6
Approval Routing	5	0	0	5
Procurement Task Creation	4	0	0	4
Email Notifications	3	0	0	3
Request Status Update	3	0	0	3
Exception Handling	4	0	0	4
Flow Execution Logs	2	0	0	2

Note: The numbers above are sample values. Replace them with the actual results from your UAT execution.

Streamlining IT Procurement: Automating Standard Laptop Orders with flow designer

Project Report Format

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1. INTRODUCTION

1.1 Project Overview

This project focuses on streamlining the IT procurement process by automating standard laptop orders using Flow Designer. The solution is designed to reduce manual effort, improve request processing, and provide a structured workflow for standard laptop procurement.

1.2 Purpose

The purpose of the project is to automate the process of requesting, reviewing, approving, and processing standard laptop orders. Flow Designer is used to connect the different stages of the procurement workflow and improve efficiency.

2. IDEATION PHASE

2.1 Problem Statement

IT laptop procurement can involve repetitive manual activities such as submitting requests, checking details, obtaining approvals, and notifying the relevant teams. These manual steps may increase processing time and create a risk of missed or inconsistent actions. The project addresses this by introducing an automated workflow for standard laptop orders.

2.2 Empathy Map Canvas

Users need a simple and reliable way to request standard laptops. They expect faster processing, clear approval status, and timely communication. IT and procurement teams need accurate request information, consistent approval handling, and reduced repetitive work.

2.3 Brainstorming

Possible improvements considered include a standardized laptop request form, automated approval routing, automatic notifications, status updates, and workflow-based task creation. Flow Designer was selected as the central automation mechanism.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey begins when an employee submits a standard laptop request. The request is reviewed and routed for approval. After approval, the required procurement or IT task is generated, relevant users are notified, and the request progresses toward completion.

3.2 Solution Requirement

The solution requires a standard laptop request form, request information fields, approval logic, Flow Designer automation, notification mechanisms, task creation, and status tracking. The workflow should handle the request consistently from submission through completion.

3.3 Data Flow Diagram

Employee → Laptop Request Form → Flow Designer → Approval → Procurement/IT Task → Notification → Request Completion.

3.4 Technology Stack

ServiceNow platform, Flow Designer, Service Catalog/request form, approval workflow, notification mechanism, and related IT service management components.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed automation directly addresses repetitive manual procurement activities by providing a standardized workflow. It helps ensure that requests follow the required approval and processing stages in a consistent manner.

4.2 Proposed Solution

A standard laptop request is submitted through the service interface. Flow Designer automatically evaluates the request, initiates the required approval, creates the next task after approval, and sends appropriate notifications. This creates a controlled end-to-end process for standard laptop orders.

4.3 Solution Architecture

The architecture consists of the user request interface, ServiceNow request records, Flow Designer automation, approval components, IT/procurement task handling, and notification services. Flow Designer acts as the orchestration layer connecting these components.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project can be planned in phases: understand the existing procurement process, identify automation requirements, design the request and approval flow, configure Flow Designer, test the workflow, resolve issues, and validate the final process.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing should verify that a laptop request can be submitted successfully, the workflow starts automatically, approvals are triggered correctly, tasks are generated after approval, notifications are sent, and the request reaches the expected completion state. The workflow should also be checked for consistent processing of standard requests.

7. RESULTS

7.1 Output Screenshots

Screenshots can be added here showing the standard laptop request form, Flow Designer workflow, approval stage, generated task, notifications, and completed request.

8. ADVANTAGES & DISADVANTAGES

Advantages: Reduces manual effort, standardizes laptop procurement, improves workflow consistency, supports faster processing, and provides better visibility of request status.

Disadvantages: Initial workflow configuration requires planning and testing. Changes to procurement rules may require workflow updates and maintenance.

9. CONCLUSION

The project demonstrates how Flow Designer can be used to automate standard laptop procurement activities. By connecting request submission, approval, task creation, and notifications into a structured workflow, the process becomes more consistent and easier to manage.

10. FUTURE SCOPE

Future improvements can include automated inventory checks, integration with procurement systems, supplier updates, advanced approval rules, automated laptop assignment, and dashboards for procurement performance and request tracking.

11. APPENDIX

Source Code (if any)

Flow Designer configuration, workflow details, and related configuration can be included here.

Dataset Link

Not applicable for the current workflow-based project.

GitHub & Project Demo Link

To be added if available.

Full Stack Development Project Documentation

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1. Introduction

- Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team ID:SWTID-2026-7461

Team Size:4

Team Leader: Selina Hepziba K

Team Members: Jansi R
Yogashree L
Thivagar M

2. Project Overview

- Purpose: To automate the standard laptop procurement request process using Flow Designer, reducing manual effort and improving request processing consistency.
- Features: Automated laptop order requests, approval workflow, task assignment, status updates, and streamlined procurement processing.

3. Architecture

- Request Interface: Users submit a standard laptop procurement request through the configured request/catalog interface.
- Flow Designer: Flow Designer manages approvals, task creation, notifications, and procurement workflow.
- Data/Records: Procurement request information, approval details, tasks, and order status are maintained in platform records.

4. Setup Instructions

- Prerequisites: ServiceNow instance access, Flow Designer, appropriate roles/permissions, and a configured standard laptop request/catalog item.
- Configuration: Configure the laptop request, approval requirements, Flow Designer workflow, actions, and test the flow in a controlled environment.

5. Workflow Structure

- Request: Employee submits a standard laptop order request.
- Approval: The request is routed to the required approver.
- Fulfilment: After approval, procurement/IT tasks are generated for processing the laptop order.
- Completion: Request status is updated after the procurement process is completed.

6. Running the Workflow

- Submit a standard laptop order request through the configured request interface.
- The Flow Designer flow starts automatically based on the configured trigger.
- Approval, task creation, notifications, and status updates are executed according to the configured flow.

7. Flow Designer Documentation

- Trigger: Standard laptop procurement request is submitted.
- Actions: Validate request details, obtain approval, create fulfilment tasks, send notifications, and update request status.
- Conditions: Approval and fulfilment actions run according to configured request conditions.

8. Access Control and Approvals

- Explain the roles and permissions required to submit, approve, and fulfil laptop procurement requests.
- Approval rules ensure requests are reviewed by the appropriate authorized personnel before fulfilment.

9. User Interface

- Provide screenshots of the laptop request form/catalog item, approval stage, Flow Designer workflow, and request status.

10. Testing

- Test standard laptop requests from submission through approval and fulfilment.
- Verify flow triggers, approval routing, task creation, notifications, status updates, and rejected/incomplete request handling.

11. Screenshots or Demo

- Provide screenshots or a demo showing the complete automated standard laptop procurement workflow in Flow Designer.

12. Known Issues

- Document any configuration, approval, notification, or fulfilment issues identified during testing.

13. Future Enhancements

- Add automated inventory validation, laptop model selection, procurement tracking, SLA monitoring, escalation notifications, and reporting/dashboard capabilities.

Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer**
- **Team ID:SWTID-2026-7461**
- **Team Size:4**
- **Team Leader: Selina Hepziba K**
- **Team Members: Jansi R
Yogashree L
Thivagar M**

2. Project Overview

- **Purpose: To automate the IT procurement process for standard laptop orders using Flow Designer, reducing manual work and improving the efficiency of laptop request and approval processes.**
- **Features:**
 - **Standard laptop request submission**
 - **Automated approval workflow**

- Request validation
- Automated task creation
- Procurement status tracking
- Notifications for request updates

3. Architecture

- **Frontend:** A simple request interface where employees can submit standard laptop requests and view their request status.
- **Backend:** Flow Designer handles the workflow, approvals, validations, task assignments, and notifications.
- **Database:** Stores laptop request details, employee information, approval status, and procurement records.

4. Setup Instructions

- **Prerequisites:** ServiceNow instance with Flow Designer access and appropriate user roles.
- **Installation:** Configure the required tables, catalog item, variables, approval rules, flow actions, notifications, and laptop procurement workflow.

5. Folder Structure

- **Request Interface:** Contains the standard laptop request form and related configurations.
- **Flow Configuration:** Contains the Flow Designer workflow, approval steps, actions, and notifications.
- **Data Configuration:** Contains request, employee, and procurement-related data.

6. Running the Application

- The employee submits a standard laptop request.
- Flow Designer automatically triggers the workflow.
- The request is validated and sent for approval.
- After approval, procurement tasks are created automatically.
- The request status is updated throughout the process.

7. API Documentation

- The project primarily uses Flow Designer actions and ServiceNow platform functionality to process requests.
- Request details are passed between workflow steps for validation, approval, task creation, and status updates.

8. Authentication

- User authentication is handled through the ServiceNow platform.
- Access to request submission, approval, and procurement activities is controlled through user roles and permissions.

9. User Interface

- Standard laptop request form
- Request details and status
- Approval information
- Procurement task status
- Automated notification messages

10. Testing

- Test standard laptop request submission.
- Verify request validation.
- Test approval and rejection scenarios.
- Verify automatic task creation.
- Check status updates and notifications.
- Test the complete workflow from request to procurement.

11. Screenshots or Demo

- Standard laptop request form
- Flow Designer workflow
- Approval step
- Procurement task
- Request status
- Notification output

12. Known Issues

- Workflow configuration may require updates when procurement policies change.
- Incorrect user roles or permissions may prevent requests or approvals from being processed.
- Notification configuration must be maintained to ensure users receive status updates.

13. Future Enhancements

- Add automatic laptop inventory checking.
- Integrate with vendor or procurement systems.
- Add automated purchase-order generation.
- Provide advanced procurement reports and dashboards.
- Add SLA-based escalation for delayed approvals.

